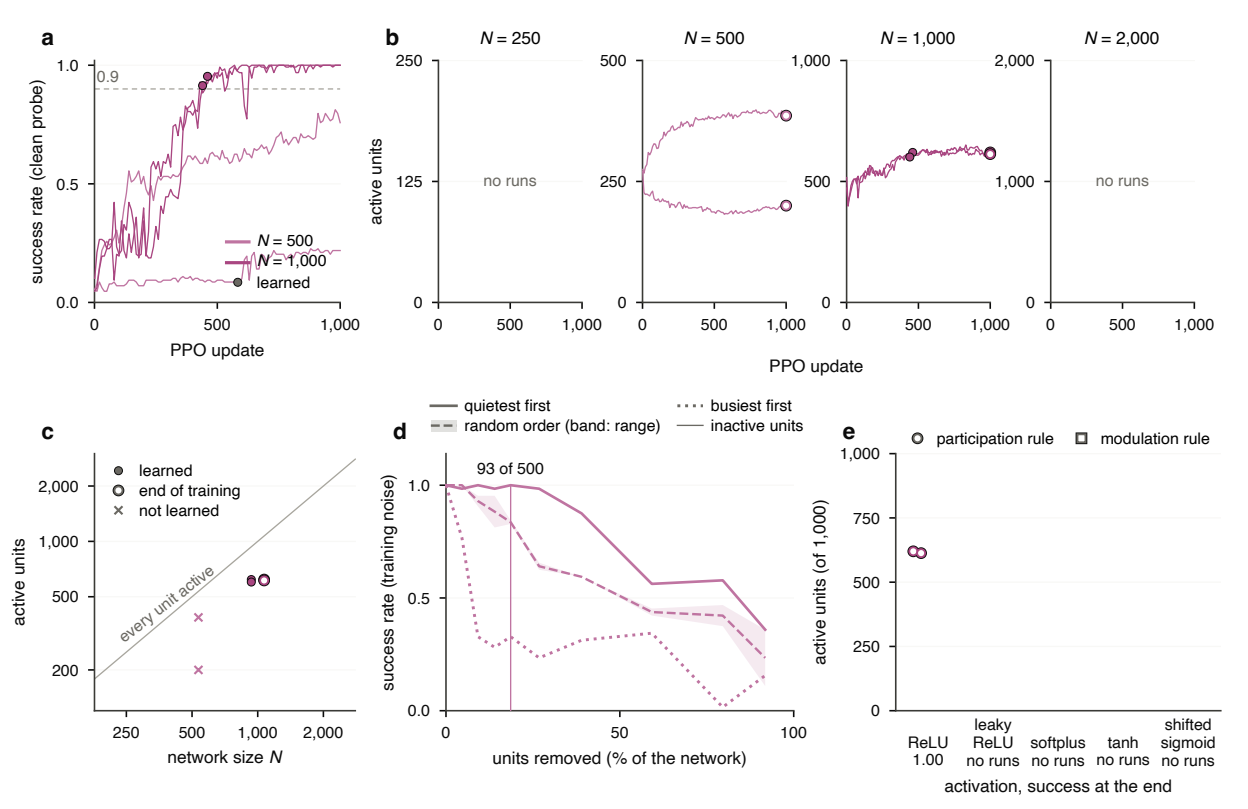
a
success rate (clean probe)
1.0
0.5
0.0
0.9
0
500
1,000
PPO update
N = 500
N = 1,000
learned
b
N = 250
N = 500
N = 1,000
N = 2,000
active units
250
125
0
no runs
500
250
1,000
500
2,000
1,000
PPO update
c
active units
2,000
1,000
500
200
learned
end of training
not learned
every unit active
250
500
1,000
2,000
network size N
d
quietest first
random order (band: range)
busiest first
inactive units
93 of 500
success rate (training noise)
1.0
0.5
0.0
0
50
100
units removed (% of the network)
e
participation rule
modulation rule
active units (of 1,000)
1,000
750
500
250
0
ReLU
1.00
leaky
ReLU
no runs
softplus
no runs
tanh
no runs
shifted
sigmoid
no runs
activation, success at the end